

PE-EC603B

Bio-Medical Electronics

Combined Module Notes

Modules 1 – 4

Semester VI • 3L:0T:0P • 3 Credits

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Disclaimer / Study Guide Notice:

These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

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Module 1

The Human Body as an Engineering System

1. The Human Body as an Engineering System

The human body can be viewed as a highly complex, self-regulating biological machine. From a bio-medical electronics perspective, it is a system with sensors (sense organs), processors (nervous system and brain), actuators (muscles), and a power supply (metabolic energy from food). Understanding this system is essential for designing instruments that measure, monitor, and assist bodily functions.

1.1 Organ Systems Overview

Definition: Organ System

An **organ system** is a group of organs that work together to perform one or more physiological functions. The human body consists of 11 major organ systems, each with specific roles in maintaining homeostasis.

Major Organ Systems Relevant to Bio-Medical Electronics:

System	Primary Organs	Clinical Relevance
Cardiovascular	Heart, arteries, veins, capillaries.	ECG, blood pressure, cardiac output monitoring.
Respiratory	Lungs, diaphragm, trachea, bronchi.	Spirometry, pulse oximetry, ventilator design.
Nervous	Brain, spinal cord, peripheral nerves.	EEG, EMG, nerve conduction velocity.
Muscular	Skeletal, smooth, cardiac muscle.	EMG, prosthetic limb control, physiotherapy.
Renal (Urinary)	Kidneys, ureter, bladder.	Dialysis machine design, fluid balance monitoring.
Endocrine	Pancreas, thyroid, adrenal glands.	Blood glucose monitoring, hormone assay.

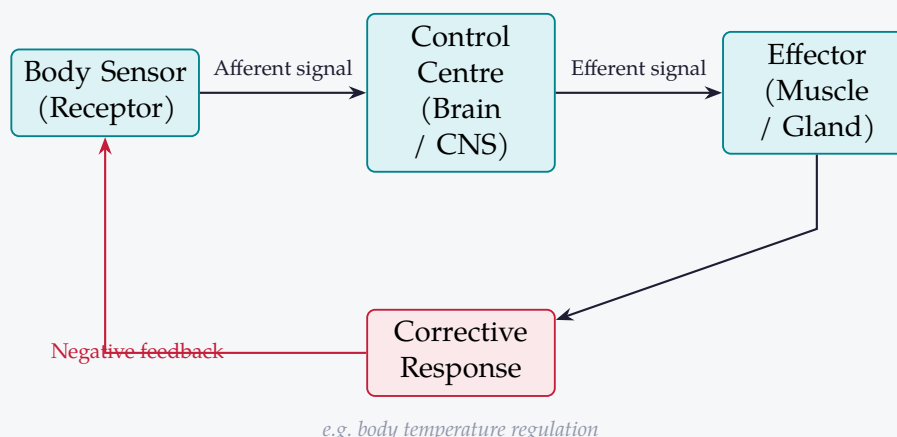
1.2 Homeostasis and Feedback Control

Definition: Homeostasis

Homeostasis is the ability of a living organism to maintain a stable internal environment despite changes in external conditions. It is achieved through continuous negative feedback loops involving sensors, control centers, and effectors.

The body maintains homeostasis by regulating parameters such as body temperature ($37 \pm 0.5^\circ\text{C}$), blood pH (7.35–7.45), blood glucose (70–100 mg/dL), and arterial blood pressure (120/80 mmHg). Most clinical instruments are designed to monitor deviations from these normal (baseline) values.

Biological Feedback Control Loop (Homeostasis)



2. The Cardiovascular System

The cardiovascular system (CVS) is the most extensively monitored system in bio-medical electronics, since the heart generates measurable electrical, pressure, and acoustic signals accessible from the body surface.

2.1 Structure and Function

Definition: Cardiovascular System

The **cardiovascular system** consists of the heart (pump), blood vessels (conduits), and blood (transport medium). It delivers oxygen, nutrients, and hormones to all tissues and removes metabolic waste products.

The heart is a four-chambered muscular pump:

- **Right Atrium (RA):** Receives deoxygenated blood from the body via the superior and inferior vena cava.
- **Right Ventricle (RV):** Pumps blood to the lungs through the pulmonary artery (pulmonary circulation).
- **Left Atrium (LA):** Receives oxygenated blood from the lungs via pulmonary veins.

- **Left Ventricle (LV):** Pumps oxygenated blood to the entire body through the aorta (systemic circulation). The LV wall is thickest because it works against the highest pressure.

2.2 The Cardiac Cycle and Electrical Conduction

Each heartbeat is initiated by the **Sinoatrial (SA) node** — the heart's natural pacemaker located in the right atrium. The electrical impulse propagates through a defined conduction pathway:

SA Node → AV Node → Bundle of His → Left and Right Bundle Branches → Purkinje Fibres → Ventricular Myocardium

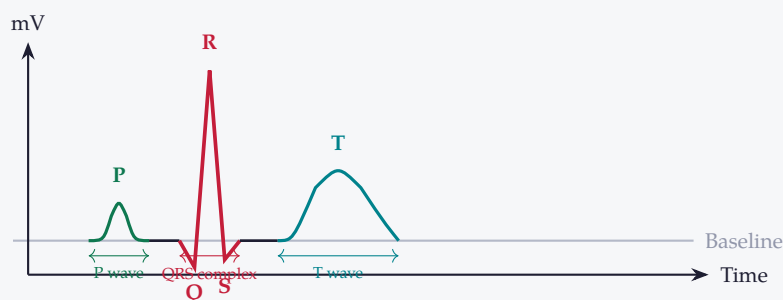
Cardiovascular Parameters and Normal Values

Heart Rate (HR)	60–100 beats/min (normal); < 60 = bradycardia; > 100 = tachycardia.
Stroke Volume (SV)	Volume ejected per beat ≈ 70 mL at rest.
Cardiac Output (CO)	$CO = HR \times SV \approx 5$ L/min at rest.
Blood Pressure (BP)	Systolic ≈ 120 mmHg; Diastolic ≈ 80 mmHg.
Mean Arterial Pressure	$MAP = \frac{SBP + 2 \times DBP}{3} \approx 93$ mmHg.

2.3 The ECG Signal

The electrocardiogram (ECG) is a recording of the electrical activity of the heart from the body surface. Each cardiac cycle produces a characteristic waveform:

ECG Waveform — Components and Significance



ECG Wave Significance:

Wave	Physiological Event	Normal Duration / Amplitude
P wave	Atrial depolarization (SA node fires, atria contract).	80–100 ms; < 0.25 mV.
PR Interval	Time from atrial to ventricular depolarization (AV node delay).	120–200 ms.
QRS Complex	Ventricular depolarization (ventricles contract).	< 120 ms; R up to 1.5 mV.
ST Segment	Ventricular plateau (all cells depolarized, isoelectric).	Flat at baseline.
T wave	Ventricular repolarization (ventricles relax).	160 ms; 0.1–0.5 mV.

3. The Respiratory System

3.1 Structure and Function

Definition: Respiratory System

The **respiratory system** is responsible for gas exchange between the body and the environment — bringing oxygen (O_2) into the blood and expelling carbon dioxide (CO_2). It consists of the conducting airways (nose, pharynx, trachea, bronchi) and the respiratory zone (alveoli where gas exchange occurs).

Mechanics of Breathing:

- **Inspiration (active):** Diaphragm and external intercostals contract → thoracic cavity volume increases → intrapulmonary pressure drops below atmospheric pressure → air flows into lungs.
- **Expiration (passive at rest):** Diaphragm relaxes → thoracic cavity recoils → pressure rises above atmospheric → air flows out.

3.2 Lung Volumes and Capacities

Lung Volume Parameters

Tidal Volume (TV)	Volume inhaled/exhaled in one normal breath ≈ 500 mL.
Inspiratory Reserve Volume (IRV)	Extra air that can be inhaled above TV ≈ 3000 mL.
Expiratory Reserve Volume (ERV)	Extra air exhaled forcefully below TV ≈ 1100 mL.
Residual Volume (RV)	Air remaining after maximal expiration ≈ 1200 mL (cannot be exhaled).
Total Lung Capacity (TLC)	$TLC = TV + IRV + ERV + RV \approx 5800$ mL.
Vital Capacity (VC)	$VC = TV + IRV + ERV \approx 4600$ mL (measurable).
FEV₁	Forced Expiratory Volume in 1 second; $FEV_1/VC > 70\%$ is normal.

3.3 Respiratory Rate and Clinical Monitoring

Normal respiratory rate (RR) is 12–20 breaths/min in adults. Clinically monitored parameters include:

- **Pulse Oximetry (SpO₂):** Non-invasive measurement of blood oxygen saturation using red and infrared LEDs. Normal SpO₂ $\geq 95\%$.
- **Capnography (EtCO₂):** Monitors exhaled CO₂ concentration, ≈ 35 –45 mmHg at end-expiration.
- **Spirometry:** Measures lung volumes and flow rates to diagnose obstructive (asthma, COPD) and restrictive disorders.

4. The Nervous System

4.1 Organisation

Definition: Nervous System

The **nervous system** is the body's primary communication and control network. It collects sensory information, processes it, and coordinates responses. It is divided into the **Central Nervous System (CNS)** — brain and spinal cord — and the **Peripheral Nervous System (PNS)** — all nerves outside the CNS.

4.2 The Neuron and Action Potential

The basic functional unit of the nervous system is the **neuron (nerve cell)**. It consists of:

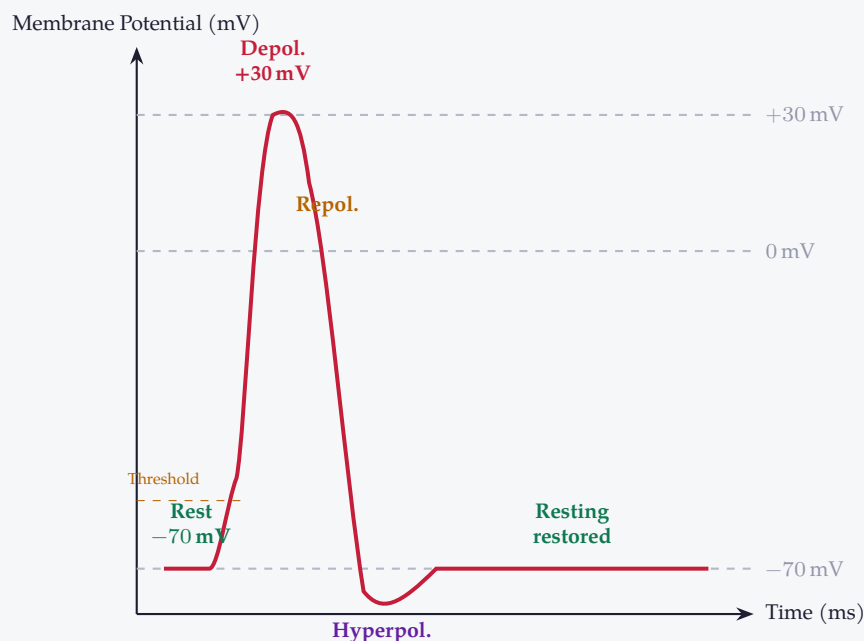
- **Dendrites:** Short branching processes that receive incoming signals from other neurons.
- **Cell Body (Soma):** Contains the nucleus and integrates incoming signals.

- **Axon:** Long fibre that carries signals (action potentials) away from the cell body to the next neuron or effector.
- **Myelin Sheath:** Insulating layer (formed by Schwann cells in PNS) that speeds up signal conduction via saltatory propagation. Conduction velocity in myelinated fibres: 40–70 m/s.

The Action Potential — Phases and Ion Movements

Resting Potential	Membrane potential ≈ -70 mV. Inside negative due to K^+ leak and Na^+/K^+ pump.
Depolarization	Na^+ channels open rapidly $\rightarrow Na^+$ rushes in $\rightarrow$ membrane potential rises to $+30$ mV.
Repolarization	Na^+ channels inactivate; K^+ channels open $\rightarrow K^+$ flows out $\rightarrow$ membrane repolarizes.
Hyperpolarization	K^+ channels close slowly $\rightarrow$ brief dip below -70 mV (refractory period).
All-or-Nothing Law	An action potential either fires fully or not at all; amplitude does not vary with stimulus strength.

Action Potential Waveform



4.3 The Brain and EEG Signals

The brain generates rhythmic electrical activity that can be recorded as an **Electroencephalogram (EEG)** from scalp electrodes. Different frequency bands correspond to different states of consciousness:

Wave	Freq. (Hz)	Amplitude (μV)	State
Delta (δ)	0.5 – 4	20 – 200	Deep sleep (slow-wave sleep).
Theta (θ)	4 – 8	20 – 100	Drowsiness, light sleep, creativity.
Alpha (α)	8 – 13	20 – 60	Relaxed wakefulness, eyes closed.
Beta (β)	13 – 30	5 – 30	Active thinking, concentration, alert.
Gamma (γ)	30 – 100	< 10	Intense cognitive processing.

5. The Muscular System and Bio-Electrical Signals Summary

Types of Muscle Tissue

- **Skeletal Muscle:** Voluntary, striated, attached to bones. Controlled by the somatic nervous system. Generates **EMG** signals.
- **Cardiac Muscle:** Involuntary, striated, forms the heart wall. Self-excitabile via SA node. Generates **ECG** signals.
- **Smooth Muscle:** Involuntary, non-striated, found in blood vessel walls, digestive tract, lungs. Controlled by the autonomic nervous system.

5.1 Summary of Bio-Electrical Signals

Signal	Physiological Origin	Amplitude	Frequency Range
ECG	Cardiac muscle depolarization/repolarization.	0.5 – 4 mV	0.05 – 150 Hz
EMG	Skeletal muscle fibre action potentials (motor units).	0.1 – 5 mV	5 – 2000 Hz
EEG	Aggregate cortical neuron activity (brain).	5 – 300 μV	0.5 – 100 Hz
EOG	Retinal dipole movement with eye motion.	0.5 – 1 mV	DC – 100 Hz
ERG	Retinal response to light stimulation.	0.5 – 1 mV	DC – 50 Hz
EGG	Gastro-intestinal smooth muscle electrical activity.	0.1 – 1 mV	0.05 – 1 Hz

Quick Revision — Module 1

Introduction to Human Physiology | PE-EC603B

Key Formulas and Normal Values

Cardiac Output	$CO = HR \times SV \quad (\approx 5 \text{ L/min at rest})$
Mean Arterial Pressure	$MAP = \frac{SBP + 2 \times DBP}{3}$
Total Lung Capacity	$TLC = TV + IRV + ERV + RV \approx 5800 \text{ mL}$
Vital Capacity	$VC = TV + IRV + ERV \approx 4600 \text{ mL}$
Nernst Equation (ion potential)	$E = \frac{RT}{zF} \ln \frac{[\text{ion}]_{\text{out}}}{[\text{ion}]_{\text{in}}}$
Resting Membrane Potential	$\approx -70 \text{ mV (inside negative)}$

One-Line Definitions

- **Homeostasis:** The body's ability to maintain a stable internal environment via negative feedback loops.
- **SA Node:** Sinoatrial node; the heart's natural pacemaker located in the right atrium; fires at 60–100 bpm.
- **Action Potential:** A brief, self-propagating reversal of membrane polarity (-70 to $+30 \text{ mV}$) that transmits nerve signals.
- **Cardiac Output:** Volume of blood pumped per minute by either ventricle; $CO = HR \times SV$.
- **Tidal Volume:** Volume of air inhaled or exhaled in one normal, quiet breath ($\approx 500 \text{ mL}$).
- **Vital Capacity:** Maximum volume of air a person can expire after a maximal inspiration.
- **Depolarization:** Rapid influx of Na^+ ions causing the cell membrane to become positive inside.
- **ECG:** Electrocardiogram; surface recording of the heart's electrical activity.
- **EEG:** Electroencephalogram; scalp recording of brain electrical activity.
- **EMG:** Electromyogram; recording of skeletal muscle electrical activity.

Frequently Asked / Viva Questions**Q1. What is homeostasis? Give an example.**

Homeostasis is the maintenance of a stable internal environment through negative feedback. Example: If body temperature rises, sweat glands are activated to cool the body back to 37 °C.

Q2. Describe the role of the SA node in the heart.

The SA node is the natural cardiac pacemaker located in the right atrium. It spontaneously depolarizes at 60–100 bpm, initiating an electrical impulse that propagates to the AV node, bundle of His, and Purkinje fibres, triggering coordinated ventricular contraction.

Q3. What physiological event does the QRS complex of the ECG represent?

The QRS complex represents ventricular depolarization — the electrical event that triggers the simultaneous contraction of the left and right ventricles.

Q4. Define cardiac output and state its normal value.

Cardiac Output (CO) is the volume of blood ejected by one ventricle per minute: $CO = HR \times SV \approx 5 \text{ L/min}$ at rest for an adult.

Q5. What is the difference between systolic and diastolic blood pressure?

Systolic pressure ($\approx 120 \text{ mmHg}$) is the peak pressure in the arteries during ventricular contraction. Diastolic pressure ($\approx 80 \text{ mmHg}$) is the minimum pressure during ventricular relaxation.

Q6. Explain the all-or-nothing law of nerve conduction.

Once a stimulus exceeds the threshold (about -55 mV), a full action potential of fixed amplitude is triggered. A sub-threshold stimulus produces no action potential at all. The amplitude does not increase with stronger stimuli.

Q7. List the EEG wave bands and their corresponding brain states.

Delta (0.5–4 Hz): deep sleep; Theta (4–8 Hz): drowsiness; Alpha (8–13 Hz): relaxed wakefulness; Beta (13–30 Hz): active alertness; Gamma ($> 30 \text{ Hz}$): intense cognition.

Q8. What is the difference between ECG, EMG, and EEG signals in terms of amplitude and frequency?

ECG: 0.5–4 mV, 0.05–150 Hz (cardiac). EMG: 0.1–5 mV, 5–2000 Hz (skeletal muscle). EEG: 5–300 μV , 0.5–100 Hz (brain) — much smaller amplitude than ECG or EMG.

Q9. What is the role of the myelin sheath in nerve conduction?

The myelin sheath acts as an electrical insulator around the axon. Depolarization jumps between nodes of Ranvier (saltatory conduction), dramatically increasing conduction velocity to 40–70 m/s compared to 0.5–2 m/s in unmyelinated fibres.

Q10. Define tidal volume and vital capacity.

Frequently Asked / Viva Questions (contd.)

Tidal Volume (TV) is the volume of air inhaled/exhaled per normal breath (≈ 500 mL). Vital Capacity (VC) is the maximum air that can be expelled after maximum inhalation ($VC = TV + IRV + ERV \approx 4600$ mL).

Q11. Why is the left ventricular wall thicker than the right?

The left ventricle pumps blood through the systemic circulation (entire body), working against a much higher resistance (pressure ≈ 120 mmHg). The right ventricle only pumps to the lungs (pulmonary pressure ≈ 25 mmHg), so it needs less force.

Bio-Electrical Signal Comparison

Signal	ECG	EMG	EEG
Source	Heart muscle	Skeletal muscle	Brain cortex
Amplitude	0.5 – 4 mV	0.1 – 5 mV	5 – 300 μ V
Bandwidth	0.05 – 150 Hz	5 – 2000 Hz	0.5 – 100 Hz
Electrodes	Surface, chest	Surface/needle	Scalp (10-20 system)
Clinical use	Arrhythmia, ischemia	Neuropathy, NCV	Epilepsy, sleep disorders

Module 2

Biomedical Transducers — Classification and Characteristics

1. Biomedical Transducers — Classification and Characteristics

Definition: Biomedical Transducer

A **biomedical transducer** is a device that converts a physiological (biological or physical) variable into an electrical signal suitable for measurement, display, recording, or processing. It acts as the primary sensing element in a medical instrument.

1.1 Classification of Biomedical Transducers

1. By Transduction Principle:

Type	Principle	Example
Resistive	Measured quantity changes resistance.	Strain gauge, thermistor, potentiometer.
Capacitive	Measured quantity changes capacitance.	Capacitive pressure sensor, microphone.
Inductive / LVDT	Movement changes mutual inductance.	LVDT (Linear Variable Differential Transformer).
Piezoelectric	Mechanical stress produces a voltage.	Ultrasonic transducer, force sensor.
Photoelectric	Light intensity changes output current.	Pulse oximeter, photodiode sensor.
Electrochemical	Ion concentration generates EMF.	pH electrode, ion-selective electrode.
Thermoelectric	Temperature difference generates voltage.	Thermocouple (Seebeck effect).

2. By Input Quantity: Displacement, Velocity, Force, Pressure, Acceleration, Flow, Temperature, Potential, Chemical concentration.

1.2 Desirable Characteristics of a Biomedical Transducer

- **High Sensitivity:** Large output per unit physiological change.
- **High Linearity:** Output proportional to input over the full range.
- **Low Hysteresis:** Output should not depend on direction of change.

- **High Signal-to-Noise Ratio (SNR):** Minimum interference and noise pick-up.
- **Biocompatibility:** Non-toxic, non-irritating, and chemically inert when implanted or in contact with tissue.
- **Minimal Loading Effect:** The transducer should not disturb the physiological system it is measuring.
- **Stability and Reproducibility:** Consistent readings over time and repeated measurements.

2. Displacement and Velocity Transducers

2.1 Strain Gauge (Resistive)

Strain Gauge

A **strain gauge** is a resistive sensor whose electrical resistance changes proportionally to the mechanical strain (deformation) applied to it. Used to measure force, pressure, torque, and displacement indirectly.

Strain Gauge Formula

$$\frac{\Delta R}{R} = G_F \cdot \varepsilon$$

where $\varepsilon = \frac{\Delta L}{L}$ is the mechanical strain, and G_F is the **Gauge Factor** (typically 2 for metal foil gauges; up to 150 for semiconductor gauges).

$$G_F = \frac{\Delta R/R}{\varepsilon} = 1 + 2\nu + \pi_\ell E$$

ν = Poisson's ratio; π_ℓ = piezoresistive coefficient; E = Young's modulus.

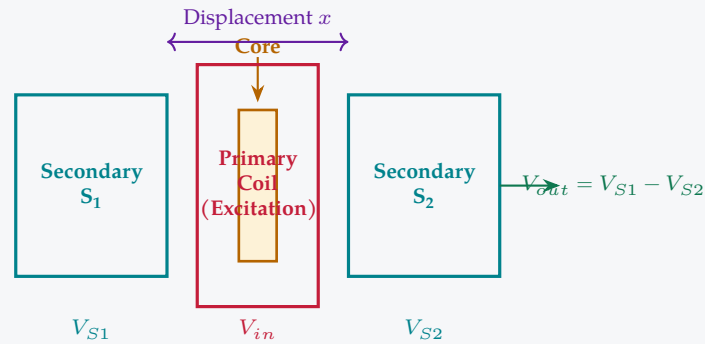
Strain gauges are arranged in a **Wheatstone bridge** to improve sensitivity and temperature compensation. For blood pressure measurement, the strain gauge is bonded to a stainless steel diaphragm.

2.2 LVDT (Linear Variable Differential Transformer)

LVDT

The **Linear Variable Differential Transformer (LVDT)** is an inductive displacement sensor that produces an AC voltage output proportional to the linear displacement of a movable ferromagnetic core.

LVDT Schematic Diagram



Working: When the core is at the centre (null position), $V_{S1} = V_{S2}$, so $V_{out} = 0$. Displacement to one side increases coupling to that secondary coil, creating a differential output proportional to displacement. The output phase indicates direction of displacement.

Biomedical Applications: Measuring chest wall movement during respiration, arterial diameter pulsation, bone growth measurement.

3. Force, Pressure, and Acceleration Transducers

3.1 Piezoelectric Transducer

Piezoelectric Effect

The **piezoelectric effect** is the generation of an electric charge on certain crystalline materials (e.g., quartz, PZT — Lead Zirconate Titanate, PVDF) when they are subjected to mechanical stress or deformation. Conversely, an applied electric field causes the material to deform (**converse piezoelectric effect**).

Piezoelectric Sensor Equations

$$Q = d_{ij} \cdot F \quad V_{out} = \frac{Q}{C_p} = \frac{d_{ij} \cdot F}{C_p}$$

where Q = generated charge (C), d_{ij} = piezoelectric charge constant (C/N), F = applied force (N), C_p = sensor capacitance (F).

$$\text{Sensitivity: } S = \frac{V_{out}}{F} = \frac{d_{ij}}{C_p} \quad (\text{V/N})$$

Biomedical Applications: Ultrasonic imaging transducers (transmit and receive), phonocardiography (heart sound recording), pulse waveform measurement, bone conduction hearing aids.

Limitations: Cannot measure static forces (DC signals bleed away through parasitic resistance). Requires high-input-impedance amplifiers.

3.2 Capacitive Pressure Sensor

A capacitive pressure sensor uses a flexible diaphragm as one plate of a capacitor. Applied pressure deflects the diaphragm, changing the gap d :

$$C = \frac{\epsilon_0 \epsilon_r A}{d} \Rightarrow \Delta C \propto \Delta P$$

Used in blood pressure catheters, intra-cranial pressure (ICP) monitoring.

3.3 Accelerometer

Accelerometer

An **accelerometer** is a transducer that measures acceleration (rate of change of velocity), typically based on a proof mass attached to a spring-damper system. Used to assess patient movement, gait analysis, and in implantable cardiac pacemakers (rate-responsive pacing).

$F = ma \Rightarrow$ deflection of mass proportional to acceleration. Piezoresistive or capacitive sensing of the deflection gives output voltage $\propto$ acceleration.

4. Flow and Temperature Transducers

4.1 Electromagnetic Blood Flow Transducer

Electromagnetic Flowmeter

Based on **Faraday's law of electromagnetic induction**: when a conductor (blood, which contains ions) moves through a magnetic field, an EMF is induced perpendicular to both the flow direction and the magnetic field.

Electromagnetic Flowmeter Formula

$$V_{emf} = B \cdot d \cdot v$$

where B = magnetic flux density (T), d = inner diameter of vessel (m), v = mean blood velocity (m/s).

$$\text{Volumetric Flow Rate: } Q = v \cdot A = v \cdot \frac{\pi d^2}{4}$$

Advantages: No obstruction to flow, direct velocity measurement, works with pulsatile flow.
Limitation: Requires surgical placement around the vessel (perivascular cuff type).

4.2 Ultrasonic (Doppler) Flow Transducer

When ultrasound is transmitted into blood and reflected from moving red blood cells, the reflected frequency is shifted:

$$\Delta f = f_r - f_t = \frac{2f_t \cdot v \cdot \cos \theta}{c}$$

where v = blood velocity, θ = angle between ultrasound beam and flow direction, c = speed of sound in tissue (≈ 1540 m/s).

Non-invasive, used in vascular Doppler studies and echocardiography.

4.3 Temperature Transducers

Type	Principle	Range	Biomedical Use
Thermistor	NTC/PTC semiconductor resistance vs. temperature.	−50 to 150 °C	Rectal/skin/blood temp.
Thermocouple	Seebeck effect: voltage at junction of dissimilar metals.	−200 to 1200 °C	Surgical probes, cryosurgery.
RTD (Pt100)	Platinum resistance increases linearly with temperature.	−200 to 600 °C	Precision lab instruments.
IC Sensor	Semiconductor bandgap vs. temperature (e.g., LM35).	−55 to 150 °C	Clinical thermometers.

Thermistor Equation (NTC):

$$R_T = R_0 \cdot e^{B\left(\frac{1}{T} - \frac{1}{T_0}\right)}$$

where B = material constant (2000–5000 K), T = temperature in Kelvin, R_0 = resistance at reference temperature T_0 .

5. Chemical and Ion-Selective Transducers

Ion-Selective Electrode (ISE)

An **ion-selective electrode** generates a potential proportional to the logarithm of the concentration of a specific ion (e.g., Na^+ , K^+ , Ca^{2+} , H^+) in the solution according to the **Nernst equation**.

Nernst Equation and pH Electrode

$$E = E^0 + \frac{RT}{zF} \ln[\text{ion}] = E^0 - \frac{0.0592}{z} \log[\text{ion}] \quad (\text{at } 25^\circ\text{C})$$

For **pH electrode** ($z = 1$ for H^+): $E = E^0 - 59.2 \text{ mV} \times \text{pH}$ (Sensitivity: 59.2 mV/pH)

Dissolved Gas Sensors:

- **Clark Electrode (pO_2):** Amperometric sensor. O_2 is reduced at a platinum cathode through a semipermeable membrane; current is proportional to pO_2 .
- **Severinghaus Electrode (pCO_2):** CO_2 diffuses through a membrane into bicarbonate solution, changing its pH, measured by a pH electrode.

6. Bio-electrodes

Bio-electrode

A **bio-electrode** is an interface between the biological medium (ionic conduction) and the measurement circuitry (electronic conduction). It converts ionic current flow in tissues to electron current flow in metallic wires.

6.1 Electrode–Electrolyte Interface and Half-Cell Potential

At the electrode–electrolyte interface, oxidation/reduction reactions create a **half-cell potential** E_{HC} . For a metal electrode:

$$M \rightleftharpoons M^{n+} + ne^{-} \quad E_{HC} = E^0 + \frac{RT}{nF} \ln[M^{n+}]$$

The **electrode offset potential** creates a DC bias that can saturate amplifiers if not accounted for.

6.2 Types of Electrodes

Type	Description	Application
Ag/AgCl Surface	Silver coated with AgCl paste; very stable, low offset potential.	Standard ECG chest electrodes, EEG.
Carbon / Paste	Conductive carbon paste; disposable, low cost.	Holter monitor, disposable ECG patches.
Needle Electrode	Fine steel or platinum needle inserted into muscle.	Clinical EMG recording, nerve conduction.
Concentric Needle	Inner wire in insulated cannula; records single motor unit.	Single motor unit EMG.
Floating Electrode	Electrode body floats; electrolyte gel bridges skin.	Long-term ambulatory monitoring.
Suction Electrode	Adheres to skin via suction (no adhesive).	Chest ECG, temporary use.
Microelectrode	Glass capillary filled with KCl; tip $< 1 \mu\text{m}$.	Intracellular recording of action potentials.

Equivalent circuit of a bio-electrode: Series combination of half-cell potential E_{HC} , electrode resistance R_e , and double-layer capacitance C_d , in series with the tissue resistance R_t .

7. Bio-potential Amplifiers and Signals**7.1 Requirements of a Bio-potential Amplifier**

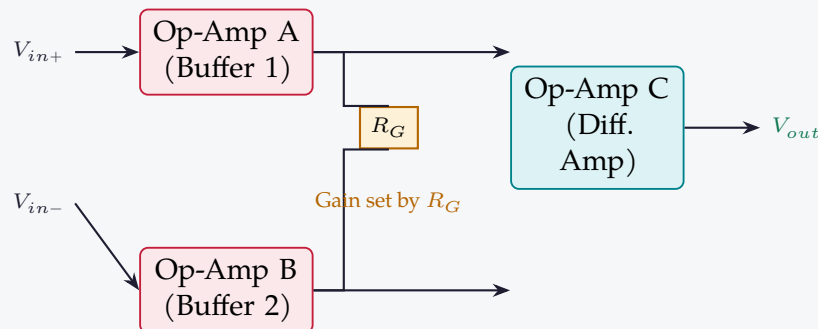
Bio-electrical signals (ECG, EMG, EEG) have very small amplitudes (μV to mV range) contaminated by electrode offset potentials, motion artefacts, and 50/60 Hz power-line interference. A bio-potential amplifier must have:

- **Very High Input Impedance** ($> 10\text{ M}\Omega$): To minimize loading of the high-impedance electrode–skin interface.
- **High Common Mode Rejection Ratio (CMRR)** ($> 80\text{ dB}$): To reject the 50 Hz power-line interference which appears as a common-mode signal on both input electrodes.
- **Low Noise**: Input-referred noise $< 1\text{ }\mu\text{V}_{rms}$ for EEG.
- **High Gain**: Sufficient to bring signals to ADC-compatible levels (typically $\times 1000$).
- **Appropriate Bandwidth**: ECG: 0.05–150 Hz; EMG: 5–2000 Hz; EEG: 0.5–100 Hz.
- **Patient Safety Isolation**: Galvanic isolation (optocouplers or transformers) to prevent micro-shock from leakage currents.

7.2 Instrumentation Amplifier (INA)

The **instrumentation amplifier (INA)** is the standard front-end for bio-potential amplifiers. It consists of three op-amps:

3-Op-Amp Instrumentation Amplifier Block Diagram



Instrumentation Amplifier Gain Formula

$$A_V = \left(1 + \frac{2R}{R_G}\right) \cdot \frac{R_f}{R_1}$$

where R_G is the single external gain-setting resistor. Increasing R_G decreases gain; $R_G \rightarrow \infty$ gives $A_V \rightarrow 1$ (unity gain).

CMRR (dB) $= 20 \log_{10} \left(\frac{A_{dm}}{A_{cm}} \right)$ Typical: 80–120 dB for medical-grade amplifiers.

7.3 ECG, EMG, and EEG Amplifier Comparison

Signal	Amplitude	Bandwidth	Gain	Key Considerations
ECG	0.5–4 mV	0.05–150 Hz	1000	High CMRR for 50 Hz rejection; right-leg drive circuit.
EMG	0.1–5 mV	5–2000 Hz	500–5000	Wide bandwidth; needle or surface electrodes.
EEG	5–300 μV	0.5–100 Hz	10 000–100 000	Extremely low noise; 19+ channel 10-20 system.

Quick Revision — Module 2

Biomedical Transducers and Bio-potential Amplifiers | PE-EC603B

Key Formulas

Strain Gauge Sensitivity	$\frac{\Delta R}{R} = G_F \cdot \varepsilon; \quad G_F \approx 2 \text{ (metal)}, \approx 100 \text{ (semiconductor)}.$
Piezoelectric Voltage	$V_{out} = \frac{d_{ij} \cdot F}{C_p}$
Thermistor (NTC)	$R_T = R_0 \cdot e^{B(1/T - 1/T_0)}$
Nernst / pH	$E = E^0 - 59.2 \times \text{pH mV at } 25^\circ\text{C}$
EM Flowmeter	$V_{emf} = B \cdot d \cdot v$
Doppler Shift	$\Delta f = \frac{2f_t v \cos \theta}{c}$
INA Gain	$A_V = \left(1 + \frac{2R}{R_G}\right) \frac{R_f}{R_1}$
CMRR	$CMRR(dB) = 20 \log_{10}(A_{dm}/A_{cm})$

One-Line Definitions

- **Transducer:** Device converting a physiological variable into an electrical signal.
- **Gauge Factor (G_F):** Ratio of relative change in resistance to applied strain.
- **LVDT:** Inductive displacement transducer whose output voltage is proportional to core displacement.
- **Piezoelectric Effect:** Generation of charge under mechanical stress in crystals like quartz or PZT.
- **Thermistor:** Semiconductor temperature sensor with a large negative (NTC) or positive (PTC) temperature coefficient of resistance.
- **Ag/AgCl Electrode:** The standard body surface bio-electrode; stable half-cell potential, low offset drift.
- **Half-cell Potential:** DC potential at the electrode-electrolyte interface due to ion exchange reactions.
- **CMRR:** Common Mode Rejection Ratio; ability of an amplifier to reject signals common to both input terminals (e.g., 50 Hz noise).
- **Instrumentation Amplifier:** Three-op-amp differential amplifier with high input impedance and high CMRR used for bio-signal acquisition.
- **Clark Electrode:** Amperometric sensor measuring dissolved oxygen partial pressure (pO_2).

Frequently Asked / Viva Questions

Q1. Define gauge factor. What is its typical value for metal and semiconductor strain gauges?

Gauge factor $G_F = (\Delta R/R)/\varepsilon$. For metal foil gauges, $G_F \approx 2$. For semiconductor (silicon) gauges, $G_F \approx 100\text{--}150$, offering much higher sensitivity.

Q2. Explain the working principle of an LVDT.

An AC voltage excites the primary coil. The ferromagnetic core couples this flux to two secondary coils differentially. Displacement of the core increases coupling to one secondary and decreases it to the other, producing an output voltage ($V_{S1} - V_{S2}$) proportional to the displacement. Phase indicates direction.

Q3. Why is the piezoelectric transducer not suitable for measuring static (DC) physiological quantities?

The piezoelectric sensor acts as a charge generator in series with a capacitance. At DC (zero frequency), charge leaks away through the amplifier input resistance, producing zero steady-state voltage. It responds only to dynamic (changing) forces.

Q4. What is the Nernst equation? What is the sensitivity of a pH glass electrode at 25 °C?

$E = E^0 - (RT/zF) \ln[\text{ion}]$. For a pH electrode (H^+ , $z = 1$), the sensitivity is 59.2 mV per pH unit at 25 °C, meaning the electrode output changes by 59.2 mV for each unit change in pH.

Q5. Why is Ag/AgCl preferred as a bio-electrode material for ECG recording?

Ag/AgCl has a very stable and reproducible half-cell potential (+0.222 V), very low polarization, and low noise. This stability minimizes electrode offset drift and motion artefacts during long-term ECG recording.

Q6. What is CMRR and why must it be high in a bio-potential amplifier?

CMRR is the ratio of differential gain to common-mode gain. The 50 Hz power-line interference appears equally on both ECG electrodes (common-mode) and can be $1000\times$ larger than the ECG signal (~ 1 mV). A CMRR of > 80 dB (10^4) reduces this interference below the signal level.

Q7. Draw the block diagram of an instrumentation amplifier and give the gain formula.

Three op-amps: two input buffers (high impedance, gain set by R_G) followed by a difference amplifier. Gain: $A_V = (1 + 2R/R_G)(R_f/R_1)$.

Q8. What is the electromagnetic flowmeter principle? Give its formula.

By Faraday's law, blood (a conductor) moving with velocity v through a magnetic field B across a vessel diameter d generates: $V_{emf} = Bvd$. Electrodes on opposite vessel walls measure this voltage.

Q9. State the Doppler effect formula for blood flow measurement.

Frequently Asked / Viva Questions (contd.)

$\Delta f = 2f_t v \cos \theta / c$. The Doppler shift Δf is proportional to blood velocity v , where f_t = transmitted frequency, θ = beam-to-flow angle, c = speed of sound in tissue.

Q10. Distinguish between surface electrodes and needle electrodes for EMG.

Surface electrodes are placed on the skin and record the combined activity of many motor units — used for general EMG and rehabilitation. Needle electrodes are inserted into the muscle belly and record individual motor unit action potentials — used in clinical diagnosis of neuropathies and myopathies.

Transducer Classification Summary

Measurand	Preferred Transducer	Output Type
Displacement	LVDT, strain gauge, potentiometer	Voltage (AC/DC)
Force / Pressure	Strain gauge bridge, piezoelectric, capacitive	Voltage / Charge
Blood Flow	EM flowmeter, Doppler ultrasound	Voltage / Freq. shift
Temperature	Thermistor (NTC), thermocouple	Resistance / Voltage
pH / Ion conc.	Ion-selective electrode (ISE)	Voltage (mV)
pO ₂ / pCO ₂	Clark / Severinghaus electrode	Current / Voltage
Bio-potential	Ag/AgCl surface, needle electrode	Voltage (μ V – mV)

Module 3

Blood Temperature Measurement

1. Blood Temperature Measurement

Body Temperature and Its Clinical Significance

Normal core body temperature is 37 °C (oral: 36.8 °C; rectal: 37.5 °C). Pyrexia (fever > 37.5 °C) and hypothermia (< 35 °C) are important clinical indicators. Core temperature is measured at the tympanic membrane, oesophagus, nasopharynx, pulmonary artery (Swan-Ganz catheter), or rectum.

1.1 Methods of Blood / Core Temperature Measurement

Method	Principle / Sensor	Site / Accuracy
Thermistor probe	NTC semiconductor; resistance drops as temperature rises.	Rectal, oesophageal, urinary bladder. Accuracy: ± 0.1 °C.
Thermocouple	Seebeck effect; voltage proportional to temperature difference between junction and reference.	Pulmonary artery catheter, cardiac surgery.
Tympanic IR	Infrared radiation from tympanic membrane detected by thermopile; proportional to temperature.	Ear canal. Fast (< 2 s), non-invasive.
Liquid crystal	Liquid crystal phase transition indicates temperature by colour change.	Forehead strips. Low accuracy (± 0.5 °C).

Thermodilution (for cardiac output): Cold saline is injected into the right atrium; a thermistor at the pulmonary artery tip detects the temperature drop as a function of time. Cardiac output is calculated using the **Stewart-Hamilton equation**:

$$CO = \frac{V_{inj}(T_B - T_{inj}) \cdot K}{\int_0^\infty \Delta T_B(t) dt}$$

where V_{inj} = injected volume, T_B = blood temperature, T_{inj} = injectate temperature, K = correction constant.

2. Blood Pressure Measurement

Blood Pressure

Blood pressure (BP) is the force exerted by circulating blood against the walls of blood vessels. Normal arterial BP is expressed as systolic/diastolic: 120/80 mmHg. It is measured directly (invasively) by intravascular catheter or indirectly (non-invasively) by sphygmomanometer.

2.1 Invasive (Direct) Blood Pressure Measurement

A fluid-filled catheter is inserted into an artery and connected to an external strain-gauge pressure transducer (or has a miniature tip-mounted sensor). The transducer converts the pressure waveform into a voltage signal displayed continuously.

Advantages: Continuous beat-to-beat waveform, accurate even in shock states.

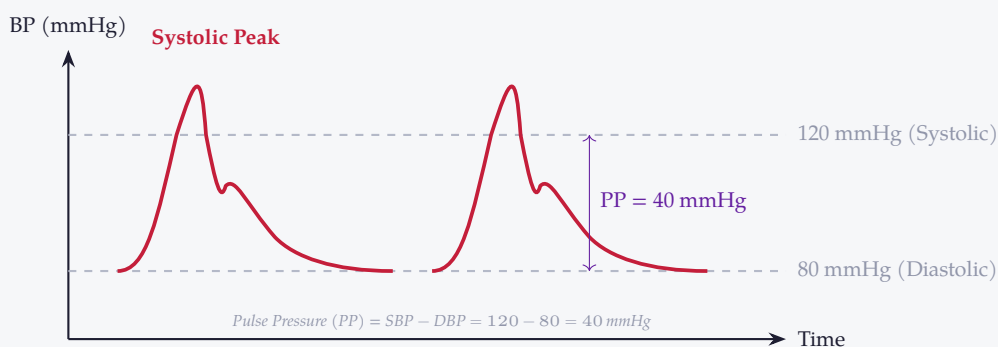
Disadvantage: Infection risk, arterial injury, thrombosis.

2.2 Non-Invasive (Indirect) Blood Pressure Measurement

Sphygmomanometer with Korotkoff Sounds Method:

1. An inflatable cuff is placed around the upper arm and inflated above systolic pressure → artery collapses, no blood flow, no sound.
2. Cuff pressure slowly deflated. When cuff pressure falls to systolic pressure, blood spurts through partially occluded artery → **first Korotkoff sound** heard (tapping). This pressure = **Systolic BP**.
3. As cuff pressure continues to fall, sounds change (tapping → swishing → dull thumping).
4. When cuff pressure drops to diastolic pressure, the artery is fully open → **sounds disappear**. This pressure = **Diastolic BP**.

Blood Pressure Waveform — Systolic, Diastolic and Pulse Pressure



Oscillometric Method (Automated NIBP monitors): The cuff is inflated and deflated automatically. A pressure sensor in the monitor detects the oscillations of the arterial wall transmitted through the cuff. Maximum oscillation occurs at **MAP** (Mean Arterial Pressure). Systolic and diastolic pressures are computed algorithmically.

Blood Pressure Formulas

Pulse Pressure $PP = SBP - DBP = 120 - 80 = 40 \text{ mmHg}$

Mean Arterial Pressure $MAP = DBP + \frac{PP}{3} = 80 + \frac{40}{3} \approx 93 \text{ mmHg}$

Alternatively $MAP = \frac{SBP + 2 \times DBP}{3}$

3. Blood Flow Measurement

3.1 Electromagnetic Flowmeter (Review)

Based on Faraday's law: $V_{emf} = Bvd$ (covered in Module 2). Used intraoperatively to measure coronary or renal artery blood flow.

3.2 Plethysmography

Plethysmography

Plethysmography is the measurement of changes in **volume** of a body part or organ, used to assess blood flow and venous function. It detects volumetric pulsations caused by arterial inflow exceeding venous outflow during systole.

Types: (a) **Strain-gauge plethysmography:** A mercury-in-silastic gauge around the limb stretches with each pulse, changing resistance $\propto$ volume change. (b) **Photo-plethysmography (PPG):** Light from an LED is transmitted through or reflected from the tissue; the varying absorption of the pulsatile blood volume modulates the photodetector output. Basis of pulse oximetry.

4. Impedance Plethysmography

Impedance Plethysmography

Impedance plethysmography measures changes in the **electrical impedance** of a body segment (typically the thorax or a limb) to infer changes in blood volume or respiratory activity. Blood has a lower resistivity than surrounding tissue, so arterial blood inflow during systole decreases the measured impedance.

4.1 Principle

A high-frequency (20–100 kHz), low-amplitude (1–4 mA rms) alternating current is injected through outer electrodes. The voltage is sensed by inner (pick-up) electrodes, and the impedance $Z = V/I$ is computed. Changes in blood volume produce small, rhythmic changes in impedance ΔZ .

4.2 Nyboer's Formula (Limb Blood Flow)

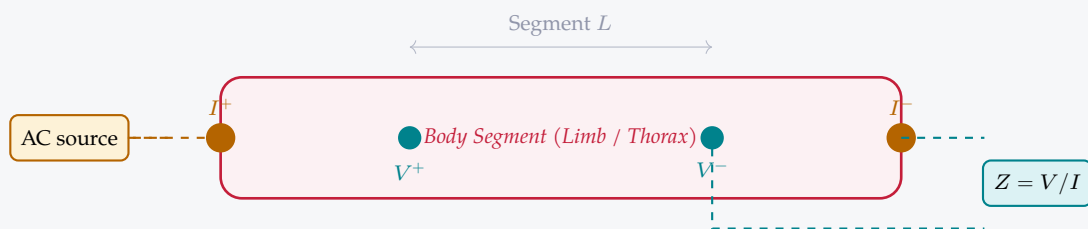
Impedance Plethysmography Formulas

$$\Delta V = \frac{\rho_b \cdot L^2}{Z_0^2} \cdot \Delta Z$$

where: ΔV = change in blood volume (mL), ρ_b = resistivity of blood ($\approx 150 \Omega \cdot \text{cm}$), L = segment length (cm), Z_0 = baseline impedance (Ω), ΔZ = impedance change (Ω).

Impedance Cardiography (ICG) measures stroke volume and cardiac output non-invasively from thoracic impedance changes.

Impedance Plethysmography — 4-Electrode Configuration



Clinical Applications of Impedance Plethysmography:

- **Cardiac output** measurement (non-invasive impedance cardiography).
- **Respiratory rate and tidal volume** monitoring (thoracic impedance changes with breathing).
- **Deep Vein Thrombosis (DVT)** detection by venous occlusion impedance plethysmography.
- **Body composition analysis** (bioelectrical impedance analysis, BIA).

5. Ultrasonic Imaging

Ultrasonic Imaging

Ultrasonic imaging uses high-frequency sound waves (1–20 MHz) generated by piezoelectric transducers to visualize internal body structures. Sound waves are reflected at tissue interfaces (impedance mismatch), and the time of return (echo) and amplitude are used to construct images.

5.1 Acoustic Impedance and Reflection

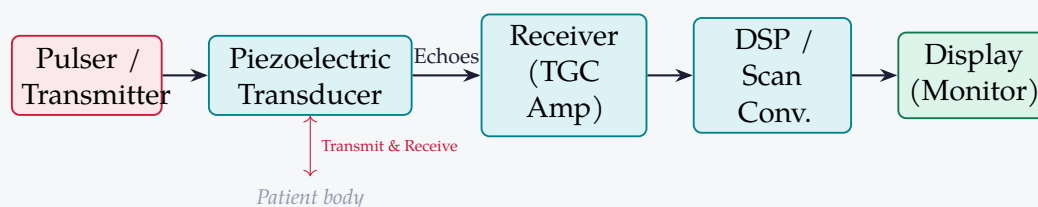
Ultrasound Key Formulas

Speed of sound in tissue	$c \approx 1540 \text{ m/s}$ (average soft tissue).
Acoustic impedance	$Z = \rho \cdot c$ (ρ = density, kg/m^3)
Reflection coefficient	$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$
Depth of reflector	$d = \frac{c \cdot t}{2}$ (t = round-trip travel time)
Axial resolution	$\Delta z = \frac{c}{2B}$ (B = pulse bandwidth)
Doppler shift	$\Delta f = \frac{2f_0 v \cos \theta}{c}$

5.2 Modes of Ultrasonic Imaging

Mode	Description	Application
A-mode (Amplitude)	Echoes displayed as amplitude spikes vs. time. 1D range measurement.	Ophthalmology (eye axial length); echoencephalography.
B-mode (Brightness)	Echo amplitude coded as brightness; 2D scan built by sweeping the beam.	Standard abdominal, obstetric ultrasound; liver, kidney, uterus.
M-mode (Motion)	Single beam; echoes displayed as position vs. time graph. Captures motion.	Echocardiography: cardiac valve motion, chamber dimension.
Doppler (Colour Flow)	Blood velocity colour-coded (red = towards, blue = away) overlaid on B-mode.	Vascular studies, cardiac valve regurgitation, stenosis.
3D / 4D	Multiple 2D planes reconstructed into 3D volume; 4D adds real-time motion.	Fetal morphology, cardiac volume assessment.

Ultrasound Imaging System Block Diagram



Time-Gain Compensation (TGC): Echoes from deeper structures are weaker due to attenuation. TGC amplifies echoes by an increasing gain as a function of time (depth) to equalize the displayed brightness across depth. Typical attenuation: $\approx 0.5 \text{ dB/cm/MHz}$.

6. X-ray Imaging

X-ray Imaging

X-rays are electromagnetic radiation (0.01–10 nm wavelength; 10^{16} – 10^{19} Hz) generated by bombarding a tungsten anode with high-velocity electrons. They penetrate tissue differentially based on tissue density and atomic number, creating a projected shadow image.

6.1 Attenuation Law

Beer-Lambert Law for X-rays

$$I = I_0 \cdot e^{-\mu x}$$

where I_0 = incident X-ray intensity, I = transmitted intensity, μ = linear attenuation coefficient (cm^{-1} , depends on material and photon energy), x = thickness of material (cm).

Tissue Contrast: Bone (μ high, appears white) > Soft tissue > Fat > Air (μ low, appears black).

6.2 Computed Tomography (CT)

Computed Tomography (CT)

CT (Computerised Tomography, or CAT scan) uses a rotating X-ray source and a detector array to acquire many projections at different angles. A reconstruction algorithm (**filtered back-projection** or iterative methods) produces cross-sectional (axial) images of the body. CT values are expressed in **Hounsfield Units (HU)**: water = 0 HU, air = −1000 HU, bone = +400 to +1000 HU.

Advantages of CT over plain X-ray: Cross-sectional images eliminate overlapping structures; better soft tissue contrast with contrast agents; enables 3D reconstruction.

Radiation dose: CT delivers a significantly higher dose than plain X-ray (chest CT $\approx$ 7 mSv vs. chest X-ray $\approx$ 0.1 mSv).

7. Nuclear Imaging

Nuclear (Radionuclide) Imaging

Nuclear imaging introduces a radioactive tracer (**radiopharmaceutical**) into the patient. The tracer accumulates in target tissues (e.g., tumour, heart). Gamma rays emitted by the tracer are detected externally to create functional images showing organ physiology and metabolism rather than anatomy alone.

7.1 Scintigraphy and Gamma Camera

A radiotracer (e.g., Technetium-99m, ^{99m}Tc , emitting 140 keV gamma rays) is injected. The **gamma camera (Anger camera)** uses a large NaI(Tl) scintillator crystal coupled to an array of photomultiplier tubes (PMTs) to detect and locate each gamma photon. A collimator ensures only perpendicularly incident photons reach the crystal. The image represents tracer distribution in the organ.

7.2 SPECT (Single Photon Emission Computed Tomography)

A gamma camera rotates around the patient, acquiring multiple projections. 3D images are reconstructed similarly to CT. Used for: myocardial perfusion imaging (heart), brain SPECT (epilepsy, dementia), bone scans.

7.3 PET (Positron Emission Tomography)

PET Scanning

PET uses positron-emitting radiotracers (e.g., ^{18}F -FDG, a glucose analogue). The emitted positron annihilates with an electron, producing two 511 keV gamma photons travelling in **opposite directions** (180°). Coincidence detection of both photons defines the line of response without a collimator, giving higher sensitivity and resolution than SPECT.

Technique	Radiation	Resolution	Key Application
X-ray	X-ray photons	0.1 mm	Bone fractures, chest pathology.
CT	X-ray photons	0.5–1 mm	Trauma, lung, abdominal, 3D anatomy.
Scintigraphy	γ rays (single)	10–15 mm	Bone scans, thyroid function.
SPECT	γ rays (single)	7–10 mm	Cardiac perfusion, brain function.
PET	γ rays (coincidence)	3–5 mm	Oncology (tumour), brain metabolism.
PET-CT	Combined	3–5 mm	Anatomy + Function fusion; gold standard in oncology.

Quick Revision — Module 3

Physiological Measurement and Medical Imaging | PE-EC603B

Key Formulas

Mean Arterial Pressure	$MAP = DBP + PP/3 = (SBP + 2 DBP)/3$
Pulse Pressure	$PP = SBP - DBP$
Thermodilution CO	$CO = \frac{V_{inj}(T_B - T_{inj}) \cdot K}{\int \Delta T_B dt}$
Impedance plethysm.	$\Delta V = \frac{\rho_b L^2}{Z_0^2} \Delta Z$
Ultrasound depth	$d = ct/2$
Acoustic impedance	$Z = \rho c$
Reflection coeff.	$R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1} \right)^2$
X-ray attenuation	$I = I_0 e^{-\mu x}$
Doppler shift	$\Delta f = 2f_0 v \cos \theta / c$

One-Line Definitions

- **Korotkoff Sounds:** Sounds heard through a stethoscope during sphygmomanometer BP measurement; first sound = systolic BP; disappearance = diastolic BP.
- **Pulse Pressure:** Difference between systolic and diastolic BP (≈ 40 mmHg).
- **Impedance Plethysmography:** Measures blood volume changes via electrical impedance changes of a body segment.
- **TGC (Time-Gain Compensation):** Increasing amplifier gain with time to compensate for ultrasound attenuation at greater depths.
- **Acoustic Impedance:** Product of tissue density and sound velocity ($Z = \rho c$); mismatch between tissues causes ultrasound reflection.
- **Hounsfield Unit:** Standardized scale of X-ray attenuation in CT; water = 0 HU, air = -1000 HU.
- **Scintigraphy:** 2D nuclear imaging using a gamma camera after radiopharmaceutical injection.
- **PET:** Positron Emission Tomography; uses annihilation coincidence detection of two 511 keV photons for 3D functional imaging.
- **A-mode ultrasound:** One-dimensional distance measurement from echo amplitude spikes on a time axis.

- **B-mode ultrasound:** 2D grey-scale image formed by brightness-coding echo amplitude as the beam is swept.

Frequently Asked / Viva Questions

Q1. Describe the Korotkoff sound method for non-invasive blood pressure measurement.

A cuff is inflated above systolic pressure (artery occluded; no sound). As cuff deflates, the first tapping Korotkoff sound marks *systolic BP*. Sounds change character until they disappear entirely at *diastolic BP*, when the artery is fully open.

Q2. What is the oscillometric method of blood pressure measurement?

An automated cuff inflates and deflates. A pressure sensor detects oscillations from the arterial wall transmitted through cuff air. Maximum oscillation amplitude occurs at MAP. Systolic and diastolic pressures are computed by algorithm from the oscillation envelope.

Q3. Define Mean Arterial Pressure and give its formula.

MAP is the average arterial pressure over one cardiac cycle, weighted for the longer diastolic phase: $MAP = DBP + PP/3 \approx 93 \text{ mmHg}$. Alternatively $MAP = (SBP + 2 \times DBP)/3$.

Q4. What is impedance plethysmography? State the Nyboer formula.

It measures blood volume changes by injecting a safe high-frequency AC current and detecting impedance changes: $\Delta V = \rho_b L^2 \Delta Z / Z_0^2$. Arterial inflow during systole decreases impedance; respiratory motion also changes thoracic impedance.

Q5. Explain acoustic impedance and its role in ultrasound imaging.

Acoustic impedance $Z = \rho c$. At the boundary between tissues with different Z , a fraction of the ultrasound energy is reflected: $R = [(Z_2 - Z_1)/(Z_2 + Z_1)]^2$. These reflections (echoes) form the image. At air–tissue interfaces, nearly all energy is reflected ($R \approx 1$), which is why coupling gel is essential.

Q6. Compare A-mode, B-mode, and M-mode ultrasound.

A-mode: amplitude spikes vs. time for 1D depth measurement (e.g., eye). B-mode: 2D brightness map used in standard diagnostic ultrasound (abdomen, obstetrics). M-mode: single beam traces target motion over time (e.g., cardiac valve motion).

Q7. State the Beer-Lambert law for X-rays and explain Hounsfield Units.

$I = I_0 e^{-\mu x}$. In CT, the Hounsfield Unit (HU) normalizes attenuation: water = 0 HU, air = -1000 HU, bone = +400 to +1000 HU. It enables standardized comparison between scanners.

Q8. What is the principle of PET scanning? How does it differ from SPECT?

Frequently Asked / Viva Questions (contd.)

PET uses positron-emitting tracers (e.g., ^{18}F -FDG). Each positron annihilates with an electron to emit two 511 keV photons at 180° . Coincidence detection defines the line of response without a collimator, giving higher sensitivity and resolution (3–5 mm). SPECT uses a single-photon emitter and a mechanical collimator, giving lower resolution (7–10 mm) but lower cost.

Q9. Why is TGC (Time-Gain Compensation) needed in ultrasound?

Ultrasound is attenuated as it travels through tissue ($\approx 0.5 \text{ dB/cm/MHz}$). Deeper echoes return later and are weaker. TGC progressively increases amplifier gain with time (depth) to compensate, so that interfaces at all depths appear equally bright on the display.

Q10. What is thermodilution and what does it measure?

Thermodilution measures cardiac output. Cold saline is injected into the right atrium; a thermistor in the pulmonary artery detects the temperature–time curve. By the Stewart-Hamilton equation, cardiac output is inversely proportional to the area under the temperature-change curve: $CO \propto 1 / \int \Delta T_B dt$.

Imaging Modality Comparison

Modality	Ionizing?	Resolution	Image type	Real-time?	Best for
X-ray	Yes	0.1 mm	2D shadow	No	Bone, chest
CT	Yes	0.5 mm	3D slices	No	Trauma, anatomy
Ultrasound	No	0.5–2 mm	2D/3D	Yes	Soft tissue, fetus
SPECT	Yes	7–10 mm	3D functional	No	Cardiac perfusion
PET	Yes	3–5 mm	3D metabolic	No	Oncology, brain
MRI	No	0.5–1 mm	3D anatomy	No	Brain, soft tissue

Module 4

Cardiac Pacemakers

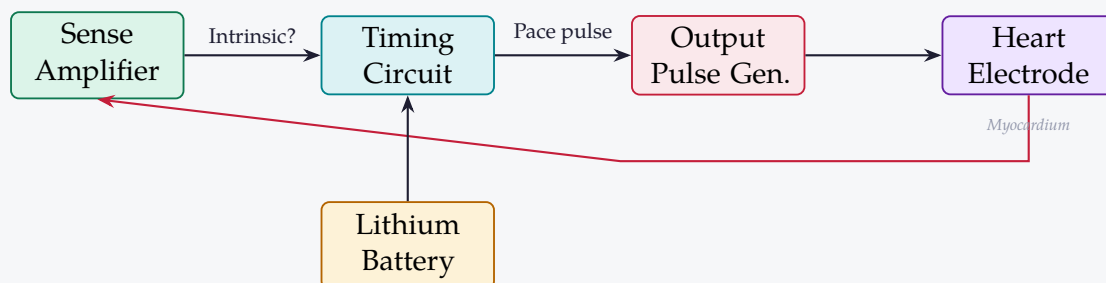
1. Cardiac Pacemakers

Definition: Cardiac Pacemaker

A **cardiac pacemaker** is an implantable electronic device that delivers controlled electrical stimuli to the heart muscle to maintain an adequate heart rate when the natural conduction system (SA node, AV node) fails or is too slow (bradycardia, heart block, sick sinus syndrome).

1.1 Pacemaker Block Diagram

Cardiac Pacemaker Block Diagram



1.2 Types of Pacemakers

Type	Description	Use
Single-chamber (AAI / VVI)	Senses and paces one chamber only (atrium or ventricle).	Sick sinus syndrome (AAI); complete heart block (VVI).
Dual-chamber (DDD)	Senses and paces both atrium and ventricle; maintains AV synchrony.	AV block with intact SA node.
Rate-responsive (DDDR)	Uses sensor (accelerometer, minute ventilation) to increase rate during exercise.	Chronotropic incompetence, active patients.
CRT (Biventricular)	Paces both ventricles simultaneously to resynchronize contraction.	Heart failure with left bundle branch block.
Leadless pacemaker	Capsule implanted directly in right ventricle; no transvenous lead.	Single-chamber pacing without lead complications.

1.3 Pacemaker Timing Parameters

Pacemaker Timing Intervals	
Pacing Rate	$\text{Rate (bpm)} = \frac{60\,000}{\text{Pacing Interval (ms)}}$
AV Delay	Time from atrial pace/sense to ventricular pace ($\approx 120\text{--}200$ ms; mimics PR interval).
Refractory Period	Time after a sensed/paced event during which the pacemaker ignores signals ($\approx 250\text{--}400$ ms).
Escape Interval	Time from a sensed beat to the next paced beat if no intrinsic beat occurs.
Pulse Width	Duration of pacing pulse ($\approx 0.4\text{--}1.0$ ms).
Pulse Amplitude	Voltage of pacing pulse ($\approx 2.5\text{--}5$ V; must exceed stimulation threshold).

Energy Source: Lithium–iodide (LiI) battery; longevity 5–15 years. Elective Replacement Indicator (ERI) detects end-of-battery voltage (2.4 V from 2.8 V nominal).

2. Defibrillators

Definition: Defibrillator

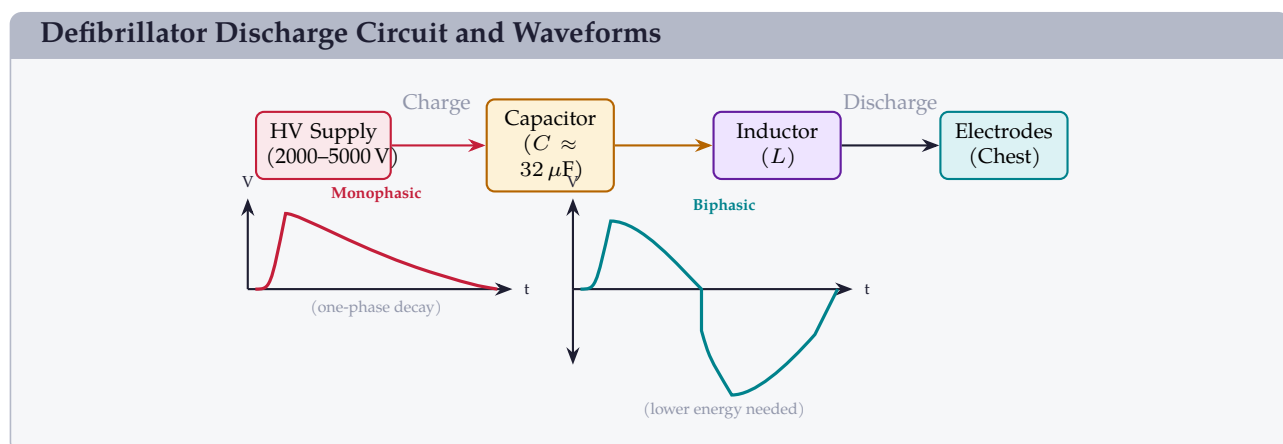
A **defibrillator** is a device that delivers a controlled, high-energy electrical shock to the heart to terminate life-threatening arrhythmias such as **ventricular fibrillation (VF)** and **ventricular tachycardia (VT)**, depolarizing all cardiac cells simultaneously so that the SA node can resume normal pacemaker control.

2.1 Principle and Types

Ventricular fibrillation is chaotic, uncoordinated cardiac electrical activity — the heart quivers rather than pumping. Without defibrillation within $\approx 4\text{--}6$ minutes, brain death ensues.

Type	Description	Energy
External	Paddle or adhesive pad electrodes on chest wall.	150–360 J (monophasic); 120–200 J (biphasic).
Internal	Used during cardiac surgery; paddles placed directly on heart.	20–50 J.
ICD (Implantable Cardioverter Defibrillator)	Permanently implanted; monitors rhythm continuously; delivers therapy automatically.	25–40 J; also provides anti-tachycardia pacing.
AED (Automated External Defibrillator)	Portable; analyses rhythm and advises/delivers shock automatically. Designed for lay-rescuer use.	120–200 J (biphasic).

2.2 Defibrillator Circuit and Waveforms



Energy stored in capacitor: $E = \frac{1}{2}CV^2$. For $C = 32 \mu\text{F}$ and $V = 3000 \text{ V}$: $E = \frac{1}{2} \times 32 \times 10^{-6} \times (3000)^2 = 144 \text{ J}$.

Biphasic vs. Monophasic: Biphasic waveforms reverse polarity partway through the shock. They are more effective at lower energies ($\approx 150\text{--}200 \text{ J}$ vs. 360 J) and cause less myocardial injury. All modern defibrillators use biphasic waveforms.

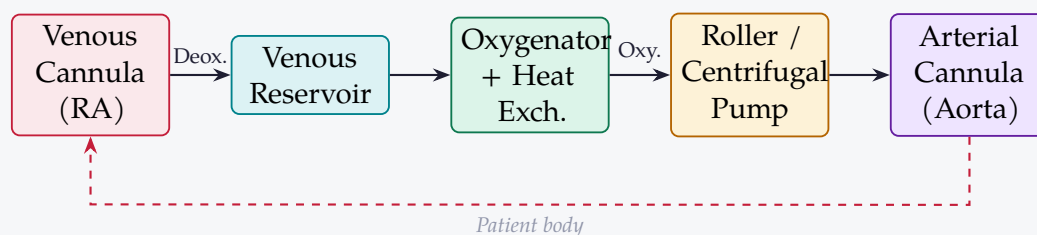
3. Heart-Lung Machine (Cardiopulmonary Bypass)

Heart-Lung Machine

The **heart-lung machine** (cardiopulmonary bypass pump, CPB) temporarily takes over the function of the heart and lungs during open-heart surgery. It oxygenates blood, removes carbon dioxide, and pumps it back to the body, allowing surgeons to operate on a still, bloodless heart.

3.1 Components and Operation

Cardiopulmonary Bypass Circuit



Key components:

- **Oxygenator (Membrane type):** Blood flows past a semipermeable membrane; O_2 diffuses in and CO_2 diffuses out. Replaces the lungs.
- **Roller Pump / Centrifugal Pump:** Provides non-pulsatile (roller) or pulsatile (centrifugal) blood flow. Replaces the heart.
- **Heat Exchanger:** Lowers body temperature (hypothermia, $28-32^\circ C$) during surgery to reduce metabolic demand; rewarms blood before weaning from bypass.
- **Arterial Filter:** Removes air bubbles and particulate emboli from the arterial return line.
- **Cardioplegia system:** Cold potassium-rich solution delivered to coronary arteries to arrest the heart in diastole (electromechanical quiescence).

4. Artificial Kidney (Dialysis)

Haemodialysis (Artificial Kidney)

Haemodialysis is a renal replacement therapy for patients with End-Stage Renal Disease (ESRD). Blood is withdrawn, passed through a **dialyser** (artificial kidney) containing a semipermeable membrane, and returned to the body. The dialyser removes metabolic waste products (urea, creatinine, excess electrolytes) and excess water by the processes of **diffusion** and **ultrafiltration**.

4.1 Principles of Dialysis

Mass Transport Mechanisms in Haemodialysis

Diffusion	Movement of solutes across membrane from high to low concentration. Main mechanism for small solutes (urea, creatinine, potassium).
Ultrafiltration	Hydraulic pressure drives water (and dissolved solutes) across the membrane. Controlled to remove excess fluid (1–3 L/session).
Clearance (K)	$K = \frac{Q_{Bi} \cdot C_{Bi} - Q_{Bo} \cdot C_{Bo}}{C_{Bi}} \quad (\text{mL/min})$
KT/V	Urea dialysis adequacy measure: $Kt/V = -\ln(R - 0.008t) + (4 - 3.5R) \times \frac{UF}{W}$ target ≥ 1.2 .

4.2 Haemodialysis Circuit

Blood is withdrawn from a vascular access (arteriovenous fistula, graft, or central venous catheter), anticoagulated with heparin, pumped through the dialyser at 200–500 mL/min and returned. The dialysate (purified water + electrolytes) flows counter-current to blood at 500–800 mL/min.

Safety alarms in a dialysis machine:

- Air bubble detector (ultrasonic).
- Venous pressure monitor (clot detection).
- Blood leak detector (photometric — detects red blood cells in dialysate).
- Conductivity monitor (ensures correct dialysate concentration).
- Temperature alarm ($\pm 1^\circ\text{C}$ from 37°C).

Peritoneal Dialysis (PD): Dialysate is instilled into the peritoneal (abdominal) cavity; the peritoneal membrane acts as the dialyser. No blood circuit required; performed at home. Less efficient than haemodialysis but better for residual renal function preservation.

5. Aids for the Handicapped

5.1 Cochlear Implant

Cochlear Implant

A **cochlear implant** is an electronic device that bypasses damaged hair cells in the cochlea and directly stimulates the auditory nerve fibres with electrical impulses, enabling profoundly deaf or severely hard-of-hearing individuals to perceive sound.

Components:

1. **Microphone** — picks up sound from environment.
2. **Speech Processor** (external, worn behind the ear) — digitizes and filters sound into frequency bands; encodes into stimulation commands.
3. **Transmitter Coil** (external) — sends encoded signals transcutaneously via radiofrequency link to the internal receiver.
4. **Internal Receiver/Stimulator** (implanted under skin) — decodes RF signal, generates electrical pulses.
5. **Electrode Array** (implanted in cochlea) — 12–22 electrodes arranged tonotopically; each electrode stimulates a different frequency region of the auditory nerve.

5.2 Prosthetic Limbs

Type	Control Mechanism	Application
Passive prosthesis	Cosmetic only; no powered movement.	Upper/lower limb amputation cosmetic restoration.
Body-powered	Harness and cable system; shoulder/elbow movement activates hook or hand.	Upper limb; robust, low cost, proprioceptive feedback.
Myoelectric (EMG-controlled)	Surface EMG from residual muscles controls motor-driven hand/elbow.	Upper limb; intuitive control; electrically powered.
Microprocessor knee	Sensors and microcontroller adjust damping in real time for variable gait.	Trans-femoral (above-knee) amputation.

5.3 Other Assistive Devices

- **Visual prosthesis (Bionic eye):** Electrode array on retina or visual cortex stimulated by camera-processed signals to restore partial vision.
- **Deep Brain Stimulator (DBS):** Electrodes implanted in subthalamic nucleus; high-frequency (130–180 Hz) electrical stimulation treats Parkinson's disease tremor.
- **Functional Electrical Stimulation (FES):** External electrodes stimulate paralysed muscles in paraplegic patients for assisted hand grasp or standing.
- **Hearing Aid:** Amplifies sound; does not bypass cochlea (unlike cochlear implant). Uses Class D amplifiers; ITE (in-the-ear) or BTE (behind-the-ear) form factors.

6. Electrical Safety in Medical Environments

Importance of Electrical Safety

Medical instruments are directly connected to or in close contact with patients. Even small leakage currents that are harmless in everyday life can be lethal in medical situations where electrodes bypass skin resistance or are in contact with the heart. Electrical safety standards (IEC 60601) define acceptable limits to protect both patients

and operators.

6.1 Macroshock and Microshock

Electric Shock Thresholds

Macroshock	Current entering through skin. Skin resistance ($\approx 10\text{ k}\Omega$ – $1\text{ M}\Omega$ dry) limits current.
Perception threshold	$\approx 1\text{ mA}$ (50 Hz AC).
Let-go threshold	≈ 10 – 20 mA (muscle tetany; cannot release conductor).
Ventricular fibrillation	$\approx 100\text{ mA}$ (chest surface current).
Cardiac arrest / Burns	≈ 1 – 10 A .
Microshock	Current applied directly to heart (via catheter, pacemaker leads). Skin resistance bypassed.
VF threshold (intracardiac)	As low as 10–50 μA (cardiac catheter, pacemaker wire).
IEC 60601 limit (patient leakage)	$\leq 10\text{ }\mu\text{A}$ (Type CF equipment connected to heart).

6.2 Sources of Leakage Current

- **Capacitive leakage:** Power line coupling through stray capacitance between mains wiring and equipment chassis.
- **Resistive leakage:** Insulation degradation due to aging, moisture, or damage.
- **Inductive leakage:** Magnetic coupling from transformers or motors.

6.3 Safety Methods and Standards

Protection Method	Description
Earth (Ground) connection	Chassis connected to protective earth; fault current diverted away from patient.
Double insulation	Two separate insulation layers between mains and chassis; no earth required.
Galvanic isolation	Patient circuit isolated from mains via transformer, optocouplers, or RF links; leakage path broken.
Right-Leg Drive (ECG)	Driven shield reduces common-mode interference; active cancellation of body surface potentials.
Equipotential bonding	All metal surfaces in the patient's area bonded to a common potential to eliminate voltage differences.
GFCI / RCD	Ground Fault Circuit Interrupter detects mains leakage $> 5 \text{ mA}$ and trips the supply.

6.4 IEC 60601 Equipment Classification

Class	Description	Max. Patient Leakage
Type B	General body contact; not directly connected to heart.	$100 \mu\text{A}$
Type BF	Floating (isolated) patient connection; body contact.	$100 \mu\text{A}$ (isolated)
Type CF	Directly connected to heart (e.g., intracardiac catheter).	$10 \mu\text{A}$

Quick Revision — Module 4

Prostheses, Therapeutic Devices and Electrical Safety | PE-EC603B

Key Formulas and Thresholds

Pacemaker rate	$\text{Rate} = 60000/\text{interval}(\text{ms})$
Defibrillator energy	$E = \frac{1}{2}CV^2 \quad (\text{J})$
Dialysis adequacy	$Kt/V \geq 1.2$ for adequate haemodialysis
Macroshock VF	$\approx 100 \text{ mA}$ (skin current, 50 Hz)
Microshock VF	$\approx 10\text{--}50 \mu\text{A}$ (intracardiac)
IEC CF leakage limit	$\leq 10 \mu\text{A}$ (direct cardiac contact)
IEC B/BF leakage limit	$\leq 100 \mu\text{A}$

One-Line Definitions

- **Pacemaker:** Implantable device delivering timed electrical pulses to maintain adequate heart rate.
- **Defibrillator:** Device delivering high-energy shock to terminate VF/VT by simultaneous depolarization of all cardiac cells.
- **AED:** Automated External Defibrillator; public-access defibrillator that analyses rhythm and advises shock delivery.
- **ICD:** Implantable Cardioverter Defibrillator; automatic implantable device sensing and treating VF/VT.
- **Heart-Lung Machine:** Cardiopulmonary bypass device replacing heart and lung function during open-heart surgery.
- **Haemodialysis:** Extracorporeal blood purification via diffusion through semipermeable membrane for renal failure.
- **Ultrafiltration:** Pressure-driven removal of water across a dialysis membrane.
- **Cochlear Implant:** Electronic device directly stimulating auditory nerve fibres to restore hearing in profound deafness.
- **Macroshock:** Dangerous electrical current entering body through skin ($> 100 \text{ mA}$ causes VF).
- **Microshock:** Tiny current ($10\text{--}50 \mu\text{A}$) directly reaching the heart via catheter/pacemaker wire; causes VF.
- **Galvanic Isolation:** Breaking the electrical connection between mains and patient circuit to eliminate leakage path.

Frequently Asked / Viva Questions

Q1. What is a cardiac pacemaker? When is it indicated?

A pacemaker is an implanted electronic device delivering controlled electrical stimuli to the heart to maintain an adequate rate. It is indicated in bradycardia, sick sinus syndrome, complete (3rd degree) AV block, and heart failure (CRT).

Q2. What are the modes of a DDD pacemaker?

DDD: first D = paces both chambers (atrium + ventricle); second D = senses both; third D = dual response (inhibited if intrinsic beat sensed, triggered if not). It maintains AV synchrony, mimicking natural conduction.

Q3. What is the difference between a pacemaker and a defibrillator?

A pacemaker delivers low-energy ($< 5\text{ V}$, $0.4\text{--}1\text{ ms}$) pulses repeatedly to sustain a minimum heart rate (treats *too slow*). A defibrillator delivers a single high-energy shock ($120\text{--}360\text{ J}$) to terminate fibrillation (treats *chaotic fast*). An ICD combines both functions.

Q4. Why are biphasic defibrillation waveforms preferred over monophasic?

Biphasic waveforms reverse polarity partway through discharge. This “tidies up” residual charge-related effects on cell membranes, achieving the same defibrillation threshold at $\approx 40\%$ lower energy, reducing post-shock myocardial injury compared to monophasic waveforms.

Q5. Explain the working of a heart-lung machine during open-heart surgery.

Venous blood is drained from the right atrium into a reservoir, passed through a membrane oxygenator (adds O_2 , removes CO_2) and a heat exchanger (induces hypothermia to reduce metabolic demand), then pumped back into the aorta. The heart is stopped with cold cardioplegia, allowing surgery on a bloodless, still field.

Q6. What is haemodialysis? Explain diffusion and ultrafiltration.

Haemodialysis removes metabolic waste from blood by passing it through a dialyser. *Diffusion*: small solutes (urea, creatinine, K^+) move down their concentration gradient across the semipermeable membrane into dialysate. *Ultrafiltration*: transmembrane pressure removes excess water from blood.

Q7. How does a cochlear implant differ from a conventional hearing aid?

A hearing aid only amplifies sound; it requires residual functional hair cells in the cochlea. A cochlear implant completely bypasses the damaged cochlea and directly stimulates auditory nerve fibres with electrical pulses, providing hearing even in profound/total deafness where no hair cells survive.

Q8. Define macroshock and microshock. Give the VF threshold for each.

Macroshock: current entering through intact skin ($> 100\text{ mA}$ at 50 Hz causes VF). Skin resistance ($10\text{ k}\Omega$ to $1\text{ M}\Omega$ dry) limits the current. *Microshock*: current reaching heart directly via catheter or pacemaker lead; VF threshold as low as $10\text{--}50\text{ }\mu\text{A}$ (skin resistance bypassed).

Frequently Asked / Viva Questions (contd.)

Q9. What is galvanic isolation and why is it essential in medical equipment?

Galvanic isolation electrically separates the patient circuit from the mains supply using a transformer, optocoupler, or RF link. Any fault current or leakage from the mains cannot reach the patient, preventing both macroshock and microshock. IEC 60601 Type CF equipment must have $\leq 10 \mu\text{A}$ patient leakage.

Q10. What is an ICD? How does it detect and treat ventricular fibrillation?

An Implantable Cardioverter Defibrillator continuously monitors the cardiac electrogram via a right ventricular lead. On detecting VF (rate > 200 bpm with disorganized waveform), it charges an internal capacitor and delivers a shock (25–40 J) within 10–20 s. It also provides anti-tachycardia pacing for slower VT to avoid a shock.

Q11. What is functional electrical stimulation (FES)?

FES applies controlled electrical pulses to surface or implanted electrodes over paralysed muscles in spinal cord injury patients. By selecting the stimulation sequence, it can activate leg muscles for assisted walking, hand muscles for grasp, or respiratory muscles for diaphragmatic pacing.

Therapeutic Device Comparison

Device	Energy Delivered	Purpose	Indication
Pacemaker	$< 5 \text{ V}, < 1 \text{ ms}$ pulses	Maintain heart rate	Bradycardia, AV block
ICD	25–40 J shock	Terminate VF/VT	VF, VT, SCD prevention
External defibrillator	120–360 J shock	Terminate VF	Cardiac arrest (VF)
CRT (BiV pacing)	Same as pacemaker	Resynchronize LV	Heart failure, LBBB
Dialysis machine	None (physical)	Remove waste, fluid	ESRD, acute renal failure
Heart-lung machine	Mechanical pump	Bypass heart/lungs	Open-heart surgery